A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-13579778})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K, the normalized generator σ_x, the pair (a', J) representing the torsion class e_j, the auxiliary divisor I', the compactly represented element t, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \mathrm{div}(t) + i_x(J) = 0, \qquad\qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 3291 & 428 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,3291 + \mathbf{Z}\,(428 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K. The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character x on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number j records which basis element e_j of $\mathrm{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x. Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2+13579778 = 0$, of discriminant -54319112. Class group invariants $(60 \quad 10 \quad 5)$, class number 3000; the three stored generator ideals (HNF) are $\begin{pmatrix} 1041 & 226 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 2103 & 1081 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1179 & 1120 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a, column e_1

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (s - 539)\, y^3 + (30s + 64173)\, y^2 + (207s + 1473998)\, y + (1786s + 8130240)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 1077*y^8 + 129424*y^7 + 16689949*y^6 + 758920670*y^5
+ 20356771897*y^4 + 397585158364*y^3 + 5253242805046*y^2 + 340088571
71352*y + 109417524002888
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{325802003}{386046315190097451} + \left(\frac{143567}{386046315190097451}\right) s, \qquad J = \begin{pmatrix} 3291 & 428 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3291$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $163263200998400 = 2^{10} \cdot 5^2 \cdot 6377468789$:

$$\begin{pmatrix}
255098751560 & 217590784500 & 81631429188 & 36234736638 & 242929058888 & 53845597857 & 204592189846 & 118214273950 & 228296114300 & 62396063166 \\
0 & 20 & 14 & 0 & 4 & 13 & 10 & 3 & 3 & 0 \\
0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 4 & 3 & 0 & 1 & 1 & 2 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 334 factors with signed exponents of absolute value up to 5380770661691368; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 0 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 2: character a, column e_2

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (s - 539)\, y^3 + (30s + 64173)\, y^2 + (207s + 1473998)\, y + (1786s + 8130240)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 1077*y^8 + 129424*y^7 + 16689949*y^6 + 758920670*y^5
+ 20356771897*y^4 + 397585158364*y^3 + 5253242805046*y^2 + 340088571
71352*y + 109417524002888
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{68485981}{7319366412146449} + \left(\frac{13914}{7319366412146449}\right) s, \qquad J = \begin{pmatrix} 1489 & 94 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1489$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $4245197219812 = 2^2 \cdot 29^2 \cdot 5189 \cdot 243197$:

$$\begin{pmatrix}
73193055514 & 60191224630 & 27811533689 & 34498423430 & 1185588565 & 45364691794 & 41037903127 & 11669585452 & 44436630101 & 39170661575 \\
0 & 29 & 17 & 13 & 10 & 6 & 8 & 17 & 2 & 23 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 339 factors with signed exponents of absolute value up to 10589744632804631; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 3: character a, column e_3

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (s - 539)\, y^3 + (30s + 64173)\, y^2 + (207s + 1473998)\, y + (1786s + 8130240)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 1077*y^8 + 129424*y^7 + 16689949*y^6 + 758920670*y^5
+ 20356771897*y^4 + 397585158364*y^3 + 5253242805046*y^2 + 340088571
71352*y + 109417524002888
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{10657907}{2278080284401899} + \Big(\frac{12625}{2278080284401899}\Big)\, s, \qquad J = \begin{pmatrix} 1179 & 1120 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1179$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2910829817600 = 2^8 \cdot 5^2 \cdot 61 \cdot 1381 \cdot 5399$:

$$\begin{pmatrix}
9096343180 & 2457015784 & 3993872520 & 3138775682 & 3129597132 & 1012554816 & 4663023730 & 6302859930 & 3281483037 & 6225601962 \\
0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\
0 & 0 & 20 & 2 & 10 & 12 & 6 & 15 & 18 & 4 \\
0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 325 factors with signed exponents of absolute value up to 5943257925666664; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 1 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 4: character b, column e_1

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 6569y^3 + 12sy^2 - 320888y - 192s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 13138*y^8 + 42509985*y^6 + 6171314576*y^4 + 40393491520*y^2 +
500604936192
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad -1 \quad -1 \quad -2 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{325802003}{386046315190097451} + \left(\frac{143567}{386046315190097451}\right) s, \qquad J = \begin{pmatrix} 3291 & 428 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3291$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $74610669539592 = 2^3 \cdot 3^6 \cdot 101 \cdot 126666581$:

$$\begin{pmatrix}
230279844258 & 136185311382 & 63265518503 & 121542316221 & 165485779116 & 29964793632 & 147546392738 & 164733312844 & 133774902679 & 152356274247 \\
0 & 6 & 0 & 3 & 0 & 0 & 3 & 3 & 1 & 4 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 3 & 0 & 0 & 1 & 0 & 0 & 2 \\
0 & 0 & 0 & 0 & 3 & 0 & 0 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 6 & 5 & 0 & 3 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 232 factors with signed exponents of absolute value up to 4546993222; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \quad 0 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 5: character b, column e_2

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 6569y^3 + 12sy^2 - 320888y - 192s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 13138*y^8 + 42509985*y^6 + 6171314576*y^4 + 40393491520*y^2 +
500604936192
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad -1 \quad -1 \quad -2 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{68485981}{7319366412146449} + (\frac{13914}{7319366412146449})\, s, \qquad J = \begin{pmatrix} 1489 & 94 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1489$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $10368548327232 = 2^6 \cdot 3^4 \cdot 131 \cdot 15267983$:

$$\begin{pmatrix}
72003807828 & 22001928450 & 16364621374 & 14955116796 & 52601643773 & 41539441693 & 35974879482 & 54114328702 & 20070849428 & 22537834490 \\
0 & 6 & 0 & 0 & 3 & 3 & 0 & 3 & 4 & 4 \\
0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 6 & 3 & 2 & 0 & 3 & 0 & 2 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 237 factors with signed exponents of absolute value up to 5236552781; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \quad 0 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 6: character b, column e_3

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 6569y^3 + 12sy^2 - 320888y - 192s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 13138*y^8 + 42509985*y^6 + 6171314576*y^4 + 40393491520*y^2 +
500604936192
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad -1 \quad -1 \quad -2 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{10657907}{2278080284401899} + (\frac{12625}{2278080284401899})\, s, \qquad J = \begin{pmatrix} 1179 & 1120 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1179$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $817477551936 = 2^6 \cdot 3^6 \cdot 131^2 \cdot 1021$:

$$\begin{pmatrix} 4815036 & 3979392 & 1105116 & 4584729 & 2463288 & 1041164 & 1823211 & 3405100 & 623830 & 4342228 \\ 0 & 6 & 0 & 3 & 3 & 3 & 0 & 3 & 4 & 1 \\ 0 & 0 & 786 & 96 & 3 & 651 & 618 & 468 & 238 & 220 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 3 & 1 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 4 & 3 & 3 & 3 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 237 factors with signed exponents of absolute value up to 2773741093; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 7: character c, column e_1

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (-s - 2221)\, y^3 + (-33s - 102794)\, y^2 + (-146s - 1892002)\, y + (2106s - 6781636)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4442*y^8 - 205588*y^7 + 14728615*y^6 + 1339313024*y^5 + 37724
552738*y^4 + 492753650160*y^3 + 3375826274932*y^2 + 17310826109888*y
+ 106220107094104
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{325802003}{386046315190097451} + (\frac{143567}{386046315190097451})\, s, \qquad J = \begin{pmatrix} 3291 & 428 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3291$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $24746154476432 = 2^4 \cdot 71 \cdot 491 \cdot 2269 \cdot 19553$:

$$\begin{pmatrix} 3093269309554 & 1941286962798 & 1473952629112 & 2734410274934 & 2484490901202 & 1901316118702 & 1838230831685 & 2324168963343 & 402940061931 & 506566045747 \\ 0 & 2 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 338 factors with signed exponents of absolute value up to 5040073251; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 4 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 8: character c, column e_2

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (-s - 2221)\, y^3 + (-33s - 102794)\, y^2 + (-146s - 1892002)\, y + (2106s - 6781636)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4442*y^8 - 205588*y^7 + 14728615*y^6 + 1339313024*y^5 + 37724
552738*y^4 + 492753650160*y^3 + 3375826274932*y^2 + 17310826109888*y
+ 106220107094104
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{68485981}{7319366412146449} + (\frac{13914}{7319366412146449})\, s, \qquad J = \begin{pmatrix} 1489 & 94 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1489$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $141442097662363 = 141442097662363$:

$$\begin{pmatrix} 141442097662363 & 58117917105825 & 34728926932359 & 24015560517198 & 56731005413023 & 15798015091734 & 89365329646601 & 101965736699696 & 120078786569384 & 21996769926247 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 337 factors with signed exponents of absolute value up to 4355775106; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 9: character c, column e_3

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (-s - 2221)\, y^3 + (-33s - 102794)\, y^2 + (-146s - 1892002)\, y + (2106s - 6781636)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4442*y^8 - 205588*y^7 + 14728615*y^6 + 1339313024*y^5 + 37724
552738*y^4 + 492753650160*y^3 + 3375826274932*y^2 + 17310826109888*y
+ 106220107094104
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{10657907}{2278080284401899} + \left(\frac{12625}{2278080284401899}\right) s, \qquad J = \begin{pmatrix} 1179 & 1120 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1179$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1204916040512 = 2^6 \cdot 179 \cdot 223 \cdot 471649$:

$$\begin{pmatrix} 75307252532 & 23924063800 & 61571877590 & 48542381732 & 14637323823 & 71620640273 & 487199243 & 31186189438 & 6515926994 & 11312771897 \\ 0 & 4 & 0 & 0 & 1 & 3 & 0 & 0 & 1 & 3 \\ 0 & 0 & 2 & 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 352 factors with signed exponents of absolute value up to 6165236700; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2956y^3 + 10sy^2 - 1077941y - 1330s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 5912*y^8 + 6582054*y^6 + 7730764992*y^4 + 800734704681*y^2 +
24021269304200
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{325802003}{386046315190097451} + \left(\frac{143567}{386046315190097451}\right) s, \qquad J = \begin{pmatrix} 3291 & 428 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3291$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $7366490174264 = 2^3 \cdot 13^2 \cdot 137 \cdot 347 \cdot 114613$:

$$\begin{pmatrix}
141663272582 & 76469067311 & 19216894240 & 96156256691 & 127578343356 & 101623736140 & 28038677140 & 61441221535 & 115269642772 & 40542442915 \\
0 & 13 & 0 & 11 & 11 & 3 & 4 & 9 & 4 & 4 \\
0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 2 & 1 & 1 & 0 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 224 factors with signed exponents of absolute value up to 6028061447; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \quad 4 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 11: character $a+b$, column e_2

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2956y^3 + 10sy^2 - 1077941y - 1330s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 5912*y^8 + 6582054*y^6 + 7730764992*y^4 + 800734704681*y^2 +
24021269304200
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{68485981}{7319366412146449} + (\frac{13914}{7319366412146449})\, s, \qquad J = \begin{pmatrix} 1489 & 94 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1489$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $5344480162400 = 2^5 \cdot 5^2 \cdot 19 \cdot 6317 \cdot 55661$:

$$\begin{pmatrix} 133612004060 & 114526395500 & 77803436137 & 32274998504 & 129862972768 & 17068873109 & 21964988053 & 39200442738 & 15138314674 & 55136504695 \\ 0 & 10 & 5 & 0 & 8 & 0 & 5 & 0 & 6 & 2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 230 factors with signed exponents of absolute value up to 18013244136; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 2 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 12: character $a+b$, column e_3

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2956y^3 + 10sy^2 - 1077941y - 1330s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 5912*y^8 + 6582054*y^6 + 7730764992*y^4 + 800734704681*y^2 +
24021269304200
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{10657907}{2278080284401899} + (\frac{12625}{2278080284401899})\, s, \qquad J = \begin{pmatrix} 1179 & 1120 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1179$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $329744801214400 = 2^6 \cdot 5^2 \cdot 11^2 \cdot 13 \cdot 19 \cdot 6895657$:

$$\begin{pmatrix}
374710001380 & 167916132076 & 169712046534 & 114610612113 & 49026677559 & 178205778244 & 154540843155 & 349796955917 & 195995086449 & 230267797779 \\
0 & 22 & 18 & 20 & 20 & 12 & 10 & 0 & 12 & 6 \\
0 & 0 & 10 & 7 & 5 & 5 & 0 & 1 & 1 & 0 \\
0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 224 factors with signed exponents of absolute value up to 10037584754; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \quad 3 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s - 1869)\, y^3 + (-26s - 33815)\, y^2 + (-203s - 99010)\, y + (-472s - 155068)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 - 3729*y^8 - 56416*y^7 + 17077809*y^6 + 832832850*y^5 +
16207803809*y^4 + 163443137484*y^3 + 913201371774*y^2 + 26330265830
56*y + 3049403346576
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0\quad 1\quad 0\quad 0\quad 0\quad 0\quad 1\quad 0\quad 0\quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{325802003}{386046315190097451} + \left(\frac{143567}{386046315190097451}\right) s, \qquad J = \begin{pmatrix} 3291 & 428 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3291$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1154658115584 = 2^{13} \cdot 3^5 \cdot 211 \cdot 2749$:

$$\begin{pmatrix}
41762808 & 38604348 & 6459228 & 39647408 & 4110197 & 7926864 & 39261976 & 7908084 & 23893939 & 27785384 \\
0 & 36 & 24 & 12 & 5 & 32 & 12 & 24 & 0 & 9 \\
0 & 0 & 12 & 6 & 4 & 4 & 0 & 6 & 9 & 7 \\
0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 4 & 0 & 0 & 3 & 3 \\
0 & 0 & 0 & 0 & 0 & 0 & 4 & 2 & 3 & 2 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 174 factors with signed exponents of absolute value up to 1351530573; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4\quad 0\quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1\quad 0\quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s - 1869)\, y^3 + (-26s - 33815)\, y^2 + (-203s - 99010)\, y + (-472s - 155068)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 - 3729*y^8 - 56416*y^7 + 17077809*y^6 + 832832850*y^5 +
16207803809*y^4 + 163443137484*y^3 + 913201371774*y^2 + 26330265830
56*y + 3049403346576
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0\quad 1\quad 0\quad 0\quad 0\quad 0\quad 1\quad 0\quad 0\quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{68485981}{7319366412146449} + (\frac{13914}{7319366412146449})\, s, \qquad J = \begin{pmatrix} 1489 & 94 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1489$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $183602455066368 = 2^8 \cdot 3^4 \cdot 13 \cdot 681098851$:

$$\begin{pmatrix} 212502841512 & 193104458724 & 54079538970 & 134308030492 & 80242678140 & 84668171960 & 161172111958 & 199038299284 & 163789615379 & 42356719491 \\ 0 & 12 & 9 & 1 & 6 & 6 & 0 & 0 & 10 & 7 \\ 0 & 0 & 3 & 2 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 6 & 2 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 170 factors with signed exponents of absolute value up to 1809898305; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \quad 2 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s - 1869)\, y^3 + (-26s - 33815)\, y^2 + (-203s - 99010)\, y + (-472s - 155068)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 - 3729*y^8 - 56416*y^7 + 17077809*y^6 + 832832850*y^5 +
16207803809*y^4 + 163443137484*y^3 + 913201371774*y^2 + 26330265830
56*y + 3049403346576
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{10657907}{2278080284401899} + (\frac{12625}{2278080284401899})\, s, \qquad J = \begin{pmatrix} 1179 & 1120 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1179$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3060658886400 = 2^8 \cdot 3^2 \cdot 5^2 \cdot 29 \cdot 1832291$:

$$\begin{pmatrix}
12752745360 & 9460680000 & 7945666181 & 6507088588 & 4471326206 & 9858787858 & 11402342686 & 11988474584 & 807606403 & 5967524832 \\
0 & 240 & 192 & 95 & 17 & 3 & 208 & 195 & 15 & 173 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 170 factors with signed exponents of absolute value up to 1099290908; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4467y^3 + 6sy^2 + 455672y - 8624s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8934*y^8 + 20865433*y^6 - 3582101640*y^4 - 1197707094080*y^2
+ 1009973935190528
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad -2 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{325802003}{386046315190097451} + \left(\frac{143567}{386046315190097451}\right) s, \qquad J = \begin{pmatrix} 3291 & 428 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3291$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $454801834496 = 2^9 \cdot 17 \cdot 52252049$:

$$\begin{pmatrix}
14212557328 & 6325238416 & 2173077432 & 4037013851 & 5732123998 & 1728825516 & 3105346657 & 1476234930 & 705963193 & 9248375407 \\
0 & 16 & 0 & 15 & 6 & 0 & 12 & 0 & 4 & 2 \\
0 & 0 & 2 & 1 & 1 & 1 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 226 factors with signed exponents of absolute value up to 389270421294; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4467y^3 + 6sy^2 + 455672y - 8624s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8934*y^8 + 20865433*y^6 - 3582101640*y^4 - 1197707094080*y^2
+ 1009973935190528
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad -2 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{68485981}{7319366412146449} + \left(\frac{13914}{7319366412146449}\right) s, \qquad J = \begin{pmatrix} 1489 & 94 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1489$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $154514518168 = 2^3 \cdot 17^5 \cdot 61 \cdot 223$:

$$\begin{pmatrix}
133663078 & 132677520 & 72787948 & 64228857 & 9538920 & 46995880 & 62054610 & 126562047 & 35847277 & 2882087 \\
0 & 34 & 0 & 28 & 16 & 30 & 28 & 10 & 18 & 0 \\
0 & 0 & 34 & 27 & 21 & 9 & 16 & 21 & 7 & 32 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 209 factors with signed exponents of absolute value up to 102535353822; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4467y^3 + 6sy^2 + 455672y - 8624s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8934*y^8 + 20865433*y^6 - 3582101640*y^4 - 1197707094080*y^2
+ 1009973935190528
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad -2 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{10657907}{2278080284401899} + \left(\frac{12625}{2278080284401899}\right) s, \qquad J = \begin{pmatrix} 1179 & 1120 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1179$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $79670475084800 = 2^{10} \cdot 5^2 \cdot 17 \cdot 19^2 \cdot 507109$:

$$\begin{pmatrix} 6551848280 & 6051694712 & 5228870644 & 501208839 & 6283881259 & 4778583955 & 4292076324 & 3192093377 & 4063646838 & 1842027053 \\ 0 & 38 & 30 & 17 & 17 & 5 & 15 & 35 & 29 & 28 \\ 0 & 0 & 4 & 1 & 1 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 & 1 & 3 & 2 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 & 2 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 209 factors with signed exponents of absolute value up to 133583081128; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.